

Master Exam Code Navigation Guide — Computer Vision Lab

Built from your 5 uploaded notebooks (you mentioned 7, but 5 were uploaded, and this guide covers exactly those 5). Every entry gives the **full notebook filename + section number + section title**, read from the notebooks' real headings. Nothing is invented. Jupyter does not show stable cell numbers, so the section number is the address.

0. HOW TO USE THIS GUIDE (read once, 60 seconds)

The five notebooks

Notebook (exact filename)	What it contains	How its sections are numbered
LAB_1_3.ipynb	"Labs 1-3 Complete Exam Code Reference": one small cell per technique, many variants of each	Section 1 = Lab 1 · Section 2 = Lab 2 · Section 3 = Lab 3 · Section 4 = combined scenarios · Section 5 = function and pitfall cheat sheet
LAB_1_3_LAB_TASKS.ipynb	"Labs 1-3 Generalized Exam Task & Scenario Codebook": big story-style templates, each with a USER SETTINGS block	1.0 to 1.9 = Lab 1 · 2.1 to 2.8 = Lab 2 · 3.1 to 3.10 = Lab 3 · 4.1 to 4.4 = combined · Part 5 = quick reference
LAB_4_5.ipynb	"Labs 4-5 Complete Exam Code Reference"	Section 2 = Lab 4 (filtering, edges, features, corners) · Section 3 = Lab 5 (segmentation) · Section 4 = combined · Section 5 = cheat sheets

Notebook (exact filename)	What it contains	How its sections are numbered
LAB_6.ipynb	"Lab 6 Complete Exam Code Reference"	Section 3 = wavelets · Section 4 = boundary / edge detection · Section 5 = Hough · Section 6 = SIFT · Section 7 = combined · Section 8 = theory
LAB_5_6_LAB_TASKS.ipynb	"Labs 5-6 Generalized Exam Task & Scenario Codebook": big story-style templates	5.1 to 5.10 = Lab 5 · 6.1 to 6.11 = Lab 6 · X.1 to X.5 = combined · Part 4 (4.1 to 4.6) = quick reference

The 5 rules

1. **Always read the notebook name together with the number.** Section numbers repeat across notebooks. Example: section 3.2.3 is "Translation without losing content" in LAB_1_3.ipynb but "Adaptive Threshold" in LAB_4_5.ipynb. And 3.7 is a 2x2-matrix transform in LAB_1_3.ipynb, Perspective Correction in LAB_1_3_LAB_TASKS.ipynb, K-Means Clustering in LAB_4_5.ipynb and the Continuous Wavelet Transform in LAB_6.ipynb.

2. Which notebook first?

- Question is a **story** ("a lab has computer screens...", "a document photographed under uneven lighting...") → open LAB_1_3_LAB_TASKS.ipynb (Labs 1-3) or LAB_5_6_LAB_TASKS.ipynb (Labs 5-6). These are the professor's tasks generalized. Each template has "If the exam question says...", "Values you may need to change" and "If it does not work immediately", and the Labs 5-6 file also has "Written-answer cheat lines" for the explain/why parts.
- Question **names a technique or gives a formula** ("apply Sobel", "log transform with c", "compute HOG", "Hough from scratch") → open LAB_1_3.ipynb, LAB_4_5.ipynb or LAB_6.ipynb (small focused cells).
- **MCQ / theory** → the cheat-sheet sections listed in section 8 of this guide.
- **Lab 4 topics** (blur and convolution, Sobel, Prewitt, LoG, Canny, HOG, LBP, corners) exist **only in LAB_4_5.ipynb** (edges also in LAB_6.ipynb section 4). There is no task-template notebook for Lab 4.

3. **How to jump:** `LAB_1_3.ipynb`, `LAB_4_5.ipynb` and `LAB_6.ipynb` have a clickable Table of Contents in their first cells. `LAB_1_3_LAB_TASKS.ipynb` and `LAB_5_6_LAB_TASKS.ipynb` have a "QUICK EXAM INDEX" in their second cell. Or press Ctrl+F and type the section number plus a word from its title, for example `3.12.3 Correct perspective`.
4. **Copy the whole cell.** Every cell is self-contained (imports, load, process, display). Change only `IMAGE_PATH` and the block labelled `USER INPUT / VALUES TO CHANGE` (in `LAB_1_3.ipynb`, `LAB_4_5.ipynb`, `LAB_6.ipynb`) or `USER SETTINGS` (in the two TASKS notebooks). If the path is wrong, the cell silently falls back to a synthetic demo image and prints a warning or "not found" message. That means: fix your path.
5. **Take the PRIMARY.** Where several places match, this guide marks **PRIMARY** (use this first), **ALTERNATIVE** (also works) and **SPECIAL CASE** (only if the question says so). Do not browse the alternatives unless the primary fails.

Which lab lives where

Lab	Topic	Notebooks and sections
1	Python and core OpenCV	<code>LAB_1_3.ipynb</code> section 1 · <code>LAB_1_3_LAB_TASKS.ipynb</code> 1.0 to 1.9
2	Photometric / point transforms (negative, log, gamma, piecewise, histogram)	<code>LAB_1_3.ipynb</code> section 2 · <code>LAB_1_3_LAB_TASKS.ipynb</code> 2.1 to 2.8
3	Geometric transforms	<code>LAB_1_3.ipynb</code> section 3 · <code>LAB_1_3_LAB_TASKS.ipynb</code> 3.1 to 3.10
4	Filtering, edges, features, corners	<code>LAB_4_5.ipynb</code> section 2 only (edges also <code>LAB_6.ipynb</code> section 4)
5	Segmentation	<code>LAB_4_5.ipynb</code> section 3 · <code>LAB_5_6_LAB_TASKS.ipynb</code> 5.1 to 5.10
6	Wavelets, boundaries, Hough, SIFT	<code>LAB_6.ipynb</code> · <code>LAB_5_6_LAB_TASKS.ipynb</code> 6.1 to 6.11

1. EMERGENCY 10-SECOND DECISION TREE

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START: what does the question WANT?
|
|-- Only load / show / save / print info / draw / label?
|   -> LAB_1_3.ipynb 1.2 (display), 1.3 (properties), 1.9 (draw), 1.15
(save)
|   story version: LAB_1_3_LAB_TASKS.ipynb 1.3, 1.4, 1.6
|
|-- Make the image look better (too dark / washed out / low contrast / hidden
detail)?
|   -> LAB 2 POINT TRANSFORMS: table 3B and guide 4.5 below
|
|-- Move / turn / resize / shear / straighten / warp the picture?
|   -> LAB 3 GEOMETRY: table 3C and guide 4.6 below
|
|-- Blur / sharpen / remove noise / apply a kernel?
|   -> LAB_1_3.ipynb 1.7 or LAB_4_5.ipynb 2.2 (noise: guide 4.7)
|
|-- Find edges / outlines / boundaries?
|   -> LAB_4_5.ipynb 2.3, LAB_6.ipynb section 4, LAB_5_6_LAB_TASKS.ipynb
5.5 (guide 4.8)
|
|-- SEPARATE something from the background (foreground / mask / cut out)?
|   -> SEGMENTATION: guide 4.1 below
|
|-- FIND / COUNT / MEASURE objects?
|   -> guide 4.2 below (LAB_5_6_LAB_TASKS.ipynb X.5 is the professor's own
version)
|
|-- Find LINES, CIRCLES, RECTANGLES, named SHAPES?
|   -> guide 4.3 below
|
|-- RECOGNIZE a known object / MATCH two images / ALIGN / PANORAMA?
|   -> guide 4.4 below
|
|-- Describe an image with numbers (HOG, LBP, histogram, texture, corners) or
compare images?
|   -> LAB_4_5.ipynb 2.4 and 2.5
|
|-- Wavelet / frequency / 1-D sensor signal / compression?
|   -> LAB_6.ipynb section 3, LAB_5_6_LAB_TASKS.ipynb 6.10 and 6.11
|

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|-- Video / frames / webcam / intruder / alarm?
|   -> LAB_1_3_LAB_TASKS.ipynb 2.8, LAB_5_6_LAB_TASKS.ipynb 6.4 / 6.6 /
6.8, LAB_6.ipynb 2.5
|
|-- Plain Python (class, dictionary, pandas table)?
|   -> LAB_1_3_LAB_TASKS.ipynb 1.1, 1.2, 1.9
|
|-- MCQ / theory / "explain why"?      -> section 8 of this guide
`-- Several steps in one question?     -> section 5 of this guide

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2. "I WANT TO..." INDEX (you know the goal, not the technique name)

I want to...	PRIMARY: notebook → section	ALTERNATIVE
Load and show an image with correct colours, handle a missing file	LAB_1_3_LAB_TASKS.ipynb → 1.3 Safe Image Loading + Correct Color Display	LAB_1_3.ipynb 1.2.1 and 1.2.2 (original vs processed)
Get size / channels / a pixel value	LAB_1_3.ipynb → 1.3.1	LAB_1_3.ipynb 1.10.2
Show the R, G or B channel	LAB_1_3.ipynb → 1.3.2	
Make it grayscale	LAB_1_3.ipynb → 1.4.1	1.4.2 (read as gray), 1.4.3 (formula)
Change colour space (HSV, LAB, YCrCb)	LAB_1_3.ipynb → 1.5.1	
Make it smaller or bigger	LAB_1_3.ipynb → 1.6 (1.6.2 exact size, 1.6.3 keep aspect ratio)	LAB_1_3.ipynb 3.4
Cut out a part of the image	LAB_1_3.ipynb → 1.8 (1.8.3 by coordinates, 1.8.4 centre, 1.8.2 quadrants)	

I want to...	PRIMARY: notebook → section	ALTERNATIVE
Write text or draw a box, circle, line	<code>LAB_1_3.ipynb</code> → 1.9	<code>LAB_1_3_LAB_TASKS.ipynb</code> 1.4 (blank canvas), 1.6 (transparent caption band)
Blur / smooth	<code>LAB_1_3.ipynb</code> → 1.7.1 Gaussian blur	<code>LAB_4_5.ipynb</code> 2.2.1 box, 2.2.2 Gaussian
Remove noise	<code>LAB_4_5.ipynb</code> → 2.2.5 Noise Removal Comparison	<code>LAB_1_3.ipynb</code> 1.17.4 (salt-and-pepper), <code>LAB_6.ipynb</code> 3.4 (wavelet)
Sharpen	<code>LAB_1_3.ipynb</code> → 1.7.4 Unsharp masking	<code>LAB_4_5.ipynb</code> 2.2.3 (kernel), <code>LAB_6.ipynb</code> 4.13.9 (Laplacian)
Fix a dark, bright or low-contrast image	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 2.1 (equalization) or 2.2 (gamma / log)	<code>LAB_1_3.ipynb</code> 2.10.1 (auto-fix)
Reveal detail hidden in dark areas	<code>LAB_1_3.ipynb</code> → 2.2 Log or 2.3 Gamma (gamma below 1)	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.2
Invert (negative)	<code>LAB_1_3.ipynb</code> → 2.1.1	
Plot or analyse a histogram	<code>LAB_1_3.ipynb</code> → 2.8	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.1
Make a colour heatmap of intensity	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 2.4 False-Color Heatmap (Colormap)	
Fix a colour cast / white balance	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 2.5 Color Balance / Color-Cast Correction	
Blend or fuse two images	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 2.7 Weighted Fusion of Two Registered Images	<code>LAB_1_3.ipynb</code> 1.13.2 <code>addWeighted</code>
Move the image	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 3.4 Translation (Shift)	<code>LAB_1_3.ipynb</code> 3.2
Rotate (keep or lose the corners)	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 3.2 Rotation Without Cropping	<code>LAB_1_3.ipynb</code> 3.3, and 1.12 (basic)

I want to...	PRIMARY: notebook → section	ALTERNATIVE
Zoom / enlarge / shrink	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 3.1 Scaling and Keep Centered	<code>LAB_1_3.ipynb</code> 3.4
Mirror / flip	<code>LAB_1_3.ipynb</code> → 3.5	<code>LAB_1_3.ipynb</code> 1.12.2
Un-slant / shear	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 3.3 Shear (Un-Slant an Image)	<code>LAB_1_3.ipynb</code> 3.6
Straighten a tilted photo or object (4 corners)	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 3.7 Perspective Correction	<code>LAB_1_3.ipynb</code> 3.12.3
Scan a document	<code>LAB_1_3.ipynb</code> → 3.12.4 (auto corners)	<code>LAB_1_3_LAB_TASKS.ipynb</code> 4.1, <code>LAB_6.ipynb</code> 7.1
Stitch a panorama	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 6.5	<code>LAB_1_3_LAB_TASKS.ipynb</code> 3.8, <code>LAB_6.ipynb</code> 6.11.3 and 6.11.4
Turn the image black and white (separate light from dark)	guide 4.1 below	
Separate by colour	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 5.4 Colour-Based Segmentation in HSV	<code>LAB_4_5.ipynb</code> 3.3.1
Find edges	<code>LAB_4_5.ipynb</code> → 2.3	<code>LAB_6.ipynb</code> section 4, <code>LAB_5_6_LAB_TASKS.ipynb</code> 5.5
Count objects	guide 4.2 below	
Measure objects (area, perimeter, centroid)	<code>LAB_5_6_LAB_TASKS.ipynb</code> → X.1	<code>LAB_4_5.ipynb</code> 4.3.2 Describe the Segmented Object
Find straight lines	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 6.1	<code>LAB_6.ipynb</code> 5.2 and 5.3
Find circles or coins	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 6.7	<code>LAB_6.ipynb</code> 5.5 and 5.7.5
Find rectangles, screens, documents	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 6.2	

I want to...	PRIMARY: notebook → section	ALTERNATIVE
Name shapes (triangle, square, circle)	LAB_6.ipynb → 7.2 Shape Detection and Counting	
Recognize a known object inside a scene	LAB_5_6_LAB_TASKS.ipynb → 6.3	LAB_6.ipynb 6.10
Compare two images	LAB_4_5.ipynb → 2.4.9 (histogram) or 4.4.1 (HOG + LBP + colour)	LAB_6.ipynb 6.11.6 (same object?)
Find corners or keypoints	LAB_4_5.ipynb → 2.5	LAB_6.ipynb section 6
Remove the background	LAB_4_5.ipynb → 3.8.1 GrabCut	LAB_4_5.ipynb 4.2.1 (by colour)
Detect an intruder or motion	LAB_5_6_LAB_TASKS.ipynb → 6.8 (video)	6.9 (two still images)
Use wavelets	LAB_6.ipynb → section 3	LAB_5_6_LAB_TASKS.ipynb 6.10, 6.11
Process a whole folder	LAB_1_3.ipynb → 1.15.2	LAB_4_5.ipynb 4.6.2, LAB_1_3.ipynb 4.0.5
Write a Python class, dictionary or pandas table	LAB_1_3_LAB_TASKS.ipynb → 1.1 / 1.2 / 1.9	
I have no idea how to segment this image	LAB_5_6_LAB_TASKS.ipynb → 5.10 (compares three methods)	LAB_4_5.ipynb 4.6.1 (tries every method)

3. "IF THE QUESTION SAYS..." MASTER TABLES (wording → technique → exact location)

This section is also the scenario dictionary: each row maps plain exam wording to the technique and to a real location. When one cell lists several sub-sections of the same notebook, they are separate cells under the same heading.

3A. Basics, drawing, statistics, plain Python (Lab 1)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
read, load, open image, display, show before and after, handle a missing file, fix colours (BGR vs RGB)	imread + None check + cvtColor + imshow	LAB_1_3_LAB_TASKS.ipynb → 1.3 Safe Image Loading + Correct Color Display	LAB_1_3.ipynb 1.2.1 display, 1.2.2 original vs processed
size, resolution, shape, number of channels, dtype, pixel value at (x, y)	img.shape, indexing	LAB_1_3.ipynb → 1.3.1	LAB_1_3.ipynb 1.10.2
extract or split the red, green, blue channel	cv2.split	LAB_1_3.ipynb → 1.3.2	
change a pixel, paint a region, make a blank black or white image	NumPy indexing, np.zeros	LAB_1_3.ipynb → 1.3.3 and 1.3.4	
draw rings, target board, concentric circles, canvas, print the centre	circle / rectangle on np.zeros	LAB_1_3_LAB_TASKS.ipynb → 1.4 Drawing Shapes on a Blank Canvas	LAB_1_3.ipynb 1.9.3
add text, caption, label, draw a rectangle on an image	putText, rectangle	LAB_1_3.ipynb → 1.9.1, and 1.9.2 (text with background box)	

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
semi-transparent box, watermark, banner with a title	addWeighted + putText	LAB_1_3_LAB_TASKS.ipynb → 1.6 Semi-Transparent Overlay Band + Text	LAB_1_3.ipynb 1.13.2
grayscale	cvtColor	LAB_1_3.ipynb → 1.4.1	1.4.2 read as gray, 1.4.3 formula, 1.4.4 back to 3 channels
convert to HSV, LAB, YCrCb, RGB	cvtColor	LAB_1_3.ipynb → 1.5.1	
resize, scale to a size, keep aspect ratio, interpolation	cv2.resize	LAB_1_3.ipynb → 1.6.1 scale factor, 1.6.2 exact size, 1.6.3 keep aspect	1.6.4 compare interpolation
crop, extract region, left or right half, quadrants, centre crop, copy a region and paste	NumPy slicing	LAB_1_3.ipynb → 1.8.1 to 1.8.5	
blur the image then crop and compare the centre ROI	GaussianBlur + slicing	LAB_1_3_LAB_TASKS.ipynb → 1.5 Blur an Image and Extract / Compare a Center ROI	LAB_1_3.ipynb 1.7 with 1.8.4
mean, min, max, std per channel, flatten channels, DataFrame, describe()	pandas	LAB_1_3_LAB_TASKS.ipynb → 1.9 Statistical Image Profiling with Pandas	LAB_1_3.ipynb 1.10.1 describe, 1.10.2 per-channel NumPy

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
percentage of dark or bright pixels, brightest or darkest pixel location	NumPy counting	LAB_1_3.ipynb → 1.17.9	
cut a shape from image 1 and put it on image 2, bitwise AND / OR / NOT, mask	bitwise ops with a 0/255 mask	LAB_1_3_LAB_TASKS.ipynb → 1.8 Masked Compositing of Two Images	LAB_1_3.ipynb 1.13.4 logo overlay with mask, 1.17.6 black-out or keep a region
add, blend, overlay, transparency of two images	addWeighted	LAB_1_3.ipynb → 1.13.2	1.13.1 cv2.add (saturated)
subtract two images, what changed, difference	absdiff	LAB_1_3.ipynb → 1.13.3	
side by side, grid, hconcat, vconcat	hconcat / vconcat	LAB_1_3.ipynb → 1.17.8	
save the result, imwrite	imwrite	LAB_1_3.ipynb → 1.15.1	
process every image in a folder	loop + glob	LAB_1_3.ipynb → 1.15.2	LAB_4_5.ipynb 4.6.2 Batch Process a Folder, LAB_1_3.ipynb 4.0.5
class with add / remove / view / total, error handling	Python class	LAB_1_3_LAB_TASKS.ipynb → 1.1 Class-Based Collection Manager	

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
nested dictionary, student records, best average, search by field	Python dict	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 1.2 Nested-Dictionary Record System	
install, import, validate libraries	pip	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 1.0 Environment Setup & Validation	<code>LAB_1_3.ipynb</code> 1.1.1

3B. Brightness, contrast, histograms (Lab 2)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
negative, invert intensities, 255 minus r	$s = 255 - r$	<code>LAB_1_3.ipynb</code> → 2.1.1	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.3 (negative is inside this cell)
log transformation, compress dynamic range, show faint structure, Fourier spectrum	$s = c \log(1+r)$	<code>LAB_1_3.ipynb</code> → 2.2.1 auto c, 2.2.2 own c, 2.2.3 inverse log, 2.2.4 why log (Fourier)	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.2 Logarithmic and Power-Law (Gamma) Transformations
gamma, power-law, gamma correction, brighten or darken	$s = c r^{\text{gamma}}$	<code>LAB_1_3.ipynb</code> → 2.3.1 formula, 2.3.2 lookup table (also colour), 2.3.3 compare gammas, 2.3.4 with constant c	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.2
plot the transformation curve T(r)	matplotlib	<code>LAB_1_3.ipynb</code> → 2.4.1	
contrast stretching, min-max, percentile stretch	linear stretch	<code>LAB_1_3.ipynb</code> → 2.5.1 min-max, 2.5.2 percentile	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.3

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
brightness and contrast, alpha and beta, $s = \alpha r + \beta$	convertScaleAbs	LAB_1_3.ipynb → 2.5.3	
piecewise linear, breakpoints, segments	np.interp + LUT	LAB_1_3.ipynb → 2.6.1 (3 segments), 2.6.2 explicit formulas, 2.6.3 N points	LAB_1_3_LAB_TASKS.ipynb 2.3 Piecewise-Linear Transform, Contrast Stretching, Negative, Intensity Slicing
highlight an intensity range, gray-level slicing, bit planes	slicing	LAB_1_3.ipynb → 2.7.1 intensity slicing, 2.7.2 bit planes	LAB_1_3_LAB_TASKS.ipynb 2.3
histogram, plot, analyse, PDF, CDF, colour histogram, compare histograms	calcHist	LAB_1_3.ipynb → 2.8.1 gray, 2.8.2 colour, 2.8.3 PDF + CDF, 2.8.4 compare	LAB_1_3_LAB_TASKS.ipynb 2.1 Histogram Analysis + Histogram Equalization
histogram equalization (cv2, from scratch, colour, CLAHE, matching)	equalizeHist, CLAHE	LAB_1_3.ipynb → 2.9.1 cv2, 2.9.2 from scratch, 2.9.3 colour, 2.9.4 CLAHE, 2.9.5 CLAHE colour, 2.9.6 matching	LAB_1_3_LAB_TASKS.ipynb 2.1 (global, CLAHE and colour in one cell); LAB_1_3.ipynb 1.14 (Lab 1 version)
dark, bright, washed-out, low-contrast, underexposed X-ray	choose by symptom	LAB_1_3.ipynb → 2.10.1 auto-fix by mean, 2.10.6 washed-out, 2.10.7 X-ray style	LAB_1_3_LAB_TASKS.ipynb 2.1 or 2.2
false colour, heatmap, applyColorMap	colormap	LAB_1_3_LAB_TASKS.ipynb → 2.4 False-Color Heatmap (Colormap)	
colour cast, white balance, gray-world, per-channel gain	channel gains	LAB_1_3_LAB_TASKS.ipynb → 2.5 Color Balance / Color-Cast Correction	
keep the brightest or densest region, mask by threshold, colour range	threshold / inRange	LAB_1_3_LAB_TASKS.ipynb → 2.6 Threshold / Mask Dense (Bright) Regions or a Color Range	LAB_1_3.ipynb 1.11, 1.17.1

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
fuse CT and MRI, two modalities, weights alpha and beta	addWeighted	LAB_1_3_LAB_TASKS.ipynb → 2.7 Weighted Fusion of Two Registered Images	LAB_1_3.ipynb 1.13.2
enhance every frame of a video, raw vs processed side by side	VideoCapture loop	LAB_1_3_LAB_TASKS.ipynb → 2.8	
apply only to a region / ROI / part of the image	mask + point transform	LAB_1_3.ipynb → 2.10.3	LAB_1_3_LAB_TASKS.ipynb 4.2 Enhance Only a Selected Region
only dark pixels, only bright pixels, conditional transform	boolean mask	LAB_1_3.ipynb → 2.10.4	
only one channel (for example brighten red)	per-channel	LAB_1_3.ipynb → 2.10.9	
report mean, std, min, max before and after; show histograms	stats + hist	LAB_1_3_LAB_TASKS.ipynb → 4.3 Multi-Step Enhancement Pipeline with Before/After Histograms and Statistics	LAB_1_3.ipynb 2.10.5, 4.0.3
compare all enhancement methods	grid	LAB_1_3.ipynb → 2.10.2	
which transformation should I use	decision helper	LAB_1_3.ipynb → 2.10.8	

3C. Geometric transformations (Lab 3)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
find pixel coordinates, pick 4 points but no numbers are given	grid / click	LAB_1_3_LAB_TASKS.ipynb → 4.4 Finding Pixel Coordinates for Source Points	LAB_1_3.ipynb 3.1.1 grid, 3.1.2 click (needs %matplotlib qt or tk), 3.1.3 mark points
translate, shift, move (for example 150 left, 80 up), 3x3 homogeneous	warpAffine	LAB_1_3_LAB_TASKS.ipynb → 3.4 Translation (Shift) with a 3x3 Homogeneous Matrix	LAB_1_3.ipynb 3.2.1 basic, 3.2.2 borders, 3.2.3 enlarged canvas
rotate by an angle, about the centre or any point, without cropping, un-tilt, sine-cosine matrix	rotation matrix	LAB_1_3_LAB_TASKS.ipynb → 3.2 Rotation Without Cropping (Manual Matrix + New Canvas Size)	LAB_1_3.ipynb 3.3.1 centre, 3.3.2 any point, 3.3.3 no cropping, 3.3.4 own matrix, 3.3.5 90/180, 1.12.1 basic
rotate then remove black corners, crop back to the original size	rotate + crop	LAB_1_3.ipynb → 3.12.6	
straighten a tilted photo (known angle)	rotation by minus the tilt	LAB_1_3_LAB_TASKS.ipynb → 3.2	LAB_1_3.ipynb 3.12.1
deskew text automatically, estimate the tilt	minAreaRect	LAB_1_3.ipynb → 3.12.2 Auto-deskew	LAB_6.ipynb 5.7.2 Deskew Using Hough
scale, enlarge, shrink, zoom, keep centred, times 3	scaling matrix	LAB_1_3_LAB_TASKS.ipynb → 3.1 Scaling (Enlarge / Shrink) and Keep Centered	LAB_1_3.ipynb 3.4.1 resize, 3.4.2 matrix, 3.4.3 zoom about centre, 3.4.4 zoom into ROI, 3.4.5 exact size
reflect, mirror, flip, symmetric image	flip / reflection matrix	LAB_1_3.ipynb → 3.5.1 flip, 3.5.2 matrix, 3.5.3 mirror image	

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
shear, skew, slant, un-slant a barcode	shear matrix	LAB_1_3_LAB_TASKS.ipynb → 3.3 Shear (Un-Slant an Image)	LAB_1_3.ipynb 3.6.1 x-shear, 3.6.2 y-shear, 3.6.3 both
apply a given 2x2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, new coordinates of points	linear transform	LAB_1_3.ipynb → 3.7.1 (auto canvas), 3.7.3 (points only)	3.7.2 anchored origin
homogeneous coordinates, compose, one master matrix, order of operations	$M = M_3 M_2 M_1$	LAB_1_3_LAB_TASKS.ipynb → 3.5 Combine Transformations into ONE Matrix (Rigid & Similarity)	LAB_1_3.ipynb 3.8.1 compose, 3.8.2 $T(c) R T(-c)$, 3.8.3 points, 3.8.4 homography on points
rigid, similarity, rotation + uniform scale + translation	rigid / similarity	LAB_1_3_LAB_TASKS.ipynb → 3.5 (MODE rigid forces scale = 1)	LAB_1_3.ipynb 3.9.1 rigid, 3.9.2 similarity, 3.9.3 estimate from point pairs
3 point pairs, solve for the matrix, landmarks, affine	getAffineTransform	LAB_1_3_LAB_TASKS.ipynb → 3.6 General Affine Matrix from 3 Point Correspondences	LAB_1_3.ipynb 3.10.1, 3.10.2 inverse
4 points, perspective, bird's-eye, rectify, make a quadrilateral rectangular, top-down view	getPerspectiveTransform	LAB_1_3_LAB_TASKS.ipynb → 3.7 Perspective Correction (Bird's-Eye View / Straighten a Planar Object)	LAB_1_3.ipynb 3.12.3 manual corners, 3.11.1
simulate a tilted view, perspective effect, affine vs perspective	warpPerspective	LAB_1_3.ipynb → 3.11.2, 3.11.3	

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
document scanner, scan a page, auto-detect corners	edges + contour + approxPolyDP + perspective	LAB_1_3.ipynb → 3.12.4 (automatic corners)	LAB_1_3_LAB_TASKS.ipynb 4.1 (corners typed in, plus binary scan); LAB_6.ipynb 7.1 (has MANUAL_POINTS option and OUT_PATH)
place an image into a frame, billboard, screen replacement	perspective into a quadrilateral	LAB_1_3_LAB_TASKS.ipynb → 3.9 Full Hierarchy: Scale, Rigid, Projective	LAB_1_3.ipynb 3.12.9
align or register two images of the same scene	ORB + homography	LAB_1_3.ipynb → 3.12.10	LAB_1_3_LAB_TASKS.ipynb 3.8, LAB_6.ipynb 6.10
stitch, panorama, homography between two photos	homography	LAB_5_6_LAB_TASKS.ipynb → 6.5 (SIFT; 6.5b is the 6-line built-in Stitcher)	LAB_1_3_LAB_TASKS.ipynb 3.8 (4 typed points, or USE_AUTO_POINTS = True with ORB); LAB_6.ipynb 6.11.3, 6.11.4
implement a warp without warpAffine, inverse mapping	manual remap	LAB_1_3_LAB_TASKS.ipynb → 3.10 Implement a Warp WITHOUT cv2.warpAffine	
where does a point go, draw the transformed point	M times point	LAB_1_3.ipynb → 3.12.8	
print matrices, pipeline scale then rotate then translate, explain the math	composition	LAB_1_3.ipynb → 3.12.11	
collage of all transformations	grid	LAB_1_3.ipynb → 3.12.7	
transform only a patch and paste it back	ROI	LAB_1_3.ipynb → 3.12.5	

3D. Filtering, noise and edges (Lab 4, Lab 6)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
blur, smooth, average, box filter	box / Gaussian	LAB_4_5.ipynb → 2.2.1 box, 2.2.2 Gaussian	LAB_1_3.ipynb 1.7.1, 1.7.2 compare blur types
apply this kernel, sharpen, emboss, outline, custom filter	filter2D	LAB_4_5.ipynb → 2.2.3 Kernel Gallery Sharpen Emboss Outline	LAB_1_3.ipynb 1.7.3, 1.7.4 unsharp masking
valid, same or stride convolution, output size	convolution	LAB_4_5.ipynb → 2.2.4 Convolution Types	
remove noise (salt-and-pepper, Gaussian)	median, Gaussian, bilateral	LAB_4_5.ipynb → 2.2.5 Noise Removal Comparison	LAB_1_3.ipynb 1.17.4 add then remove, 1.7.2; LAB_6.ipynb 3.4 wavelet denoise, 4.13.1 edges in a noisy image
Sobel, gradient magnitude, Gx and Gy, vertical or horizontal edges	Sobel	LAB_4_5.ipynb → 2.3.1	LAB_6.ipynb 4.1 Sobel, 4.2 variations, 4.3 binary edge map, 4.4 kernel by hand
Prewitt, Roberts, Scharr	first-derivative kernels	LAB_4_5.ipynb → 2.3.3 Prewitt, 2.3.2 Scharr	LAB_6.ipynb 4.5 (all three)
Laplacian	second derivative	LAB_4_5.ipynb → 2.3.4	LAB_6.ipynb 4.9, 4.13.9 sharpen with Laplacian
LoG, Marr-Hildreth, zero crossings, DoG	Laplacian of Gaussian	LAB_4_5.ipynb → 2.3.5	LAB_6.ipynb 4.9, 4.10 zero crossings, 4.11 kernel and DoG

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
Canny, change thresholds, too many or too few edges, hysteresis	Canny	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 5.5 Canny Edge Detection with Hysteresis Threshold Study	<code>LAB_4_5.ipynb</code> 2.3.6 and 3.4.1; <code>LAB_6.ipynb</code> 4.6 manual code, 4.7 threshold experiments; <code>LAB_1_3.ipynb</code> 1.17.7
edges without choosing thresholds, automatic Canny	median-based thresholds	<code>LAB_4_5.ipynb</code> → 2.3.7 Auto Canny	<code>LAB_6.ipynb</code> 4.8
compare edge detectors	side by side	<code>LAB_6.ipynb</code> → 4.12 Compare Sobel Canny LoG	
overlay or highlight edges on the original	overlay	<code>LAB_6.ipynb</code> → 4.13.2	
only horizontal or only vertical edges	Sobel x or y	<code>LAB_6.ipynb</code> → 4.13.3	
edges per colour channel	per-channel Canny	<code>LAB_6.ipynb</code> → 4.13.4	
how many edge pixels, edge density, dominant edge direction	statistics	<code>LAB_6.ipynb</code> → 4.13.7	<code>LAB_4_5.ipynb</code> 4.4.2
edges only inside a region	ROI	<code>LAB_6.ipynb</code> → 4.13.8	
boundary of a binary region (morphological)	dilate minus erode	<code>LAB_6.ipynb</code> → 4.13.6	
contours, outline, object boundary	findContours	<code>LAB_6.ipynb</code> → 4.13.5	<code>LAB_4_5.ipynb</code> 3.4.3

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
save edge results	imwrite	LAB_6.ipynb → 4.13.10	

3E. Segmentation, counting, measuring (Lab 5)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
threshold at a value, binary, invert, truncate, all threshold types	cv2.threshold	LAB_4_5.ipynb → 3.2.1 Global Threshold All Types	LAB_1_3.ipynb 1.11.1 binary, 1.11.2 all types
automatic threshold, Otsu, bimodal histogram, objects on a plain background	Otsu	LAB_5_6_LAB_TASKS.ipynb → 5.3 Otsu Automatic Thresholding with Histogram and Stability Check	LAB_4_5.ipynb 3.2.2; LAB_1_3.ipynb 1.11.3
uneven lighting, shadow, receipt, text on paper, adaptive, compare global vs adaptive vs Otsu	adaptiveThreshold	LAB_5_6_LAB_TASKS.ipynb → 5.1 Global vs Adaptive vs Otsu Threshold Comparison	LAB_5_6_LAB_TASKS.ipynb 5.2 (parameter grid: block size x C x method); LAB_4_5.ipynb 3.2.3 and 4.5.1; LAB_1_3_LAB_TASKS.ipynb 1.7 (global vs adaptive, then rotate the result)
segment by colour (red, yellow...), HSV, inRange, lower and upper bound, extract a colour	HSV mask	LAB_5_6_LAB_TASKS.ipynb → 5.4 Colour-Based Segmentation in HSV	LAB_4_5.ipynb 3.3.1, 3.3.2 find the HSV of a pixel, 3.3.3 BGR thresholding; LAB_1_3.ipynb 1.17.1
extract an object by colour and replace or change the background	mask + bitwise_and	LAB_4_5.ipynb → 4.2.1	
segment using edges, edges to a filled mask	Canny + closing + fill	LAB_4_5.ipynb → 3.4.2	

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
region growing, seed point, contiguous similar region, MRI region	seeded growing	LAB_5_6_LAB_TASKS.ipynb → 5.6 Region Growing from Seed Points	LAB_4_5.ipynb 3.5.1 manual, 3.5.2 floodFill
touching objects, watershed, sure foreground, distance transform, markers, separate coins	marker watershed	LAB_5_6_LAB_TASKS.ipynb → 5.7 Marker-Based Watershed	LAB_4_5.ipynb 3.6.1; tuning table: LAB_5_6_LAB_TASKS.ipynb 5.8
K-Means, clusters, reduce colours, quantisation, K = 2, 4, 6	cv2.kmeans	LAB_5_6_LAB_TASKS.ipynb → 5.9 K-Means Colour Clustering Segmentation	LAB_4_5.ipynb 3.7.1 colour, 3.7.2 gray, 3.7.3 with position, 3.7.4 choose K (elbow), 3.7.5 mean shift
isolate one cluster, then count or extract it	kmeans + contours	LAB_5_6_LAB_TASKS.ipynb → X.3	
remove the background, foreground extraction, GrabCut	grabCut	LAB_4_5.ipynb → 3.8.1	
superpixels, graph-based / active contour, snake / deep-learning segmentation	specialised	LAB_4_5.ipynb → 3.8.2 / 3.8.3 / 3.8.4 (3.8.4 needs torch: skip unless installed)	
clean the mask, remove small specks, fill holes, morphology, opening, closing, erode, dilate	morphology	LAB_4_5.ipynb → 3.9.1 Erode Dilate Open Close	morphology is also built into LAB_5_6_LAB_TASKS.ipynb 5.7, X.1 and 5.4 through OPEN_ITER and CLOSE_ITER

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
choose or compare three methods, unknown image, justify your choice	method comparison	LAB_5_6_LAB_TASKS.ipynb → 5.10	LAB_4_5.ipynb 4.6.1 Try Every Segmentation Method Side by Side
crop the largest object	contours + max area	LAB_4_5.ipynb → 4.2.2	
blur the background, filter only the object, operate only on the segmented region	mask	LAB_4_5.ipynb → 4.3.1	
area, perimeter, centroid, bounding box of each object	contour measurements	LAB_5_6_LAB_TASKS.ipynb → X.1 Detect, Count, Measure and Extract Individual Objects	LAB_4_5.ipynb 4.3.2 Describe the Segmented Object
count, how many objects, number of separate regions	see guide 4.2 below		

3F. Lines, circles, rectangles, shapes (Lab 6)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
Hough concept, slope-intercept, parameter space, voting, accumulator	theory	LAB_6.ipynb → 5.1	

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
detect straight lines, boundaries, screen edges, horizontal or vertical classification	HoughLines / HoughLinesP	LAB_5_6_LAB_TASKS.ipynb → 6.1 Detect Straight Lines with the Hough Line Transform	LAB_6.ipynb 5.2 HoughLines (infinite lines), 5.3 HoughLinesP (segments)
Hough from scratch, implement the accumulator	manual voting	LAB_6.ipynb → 5.4 lines, 5.6 circles	
detect or count circles, coins, radius groups, crop circles	HoughCircles	LAB_5_6_LAB_TASKS.ipynb → 6.7 Detect and Count Circular Objects with the Hough Circle Transform	LAB_6.ipynb 5.5, 5.7.5
lane, road lines, vehicle camera	Canny + ROI + Hough	LAB_5_6_LAB_TASKS.ipynb → 6.6	LAB_6.ipynb 5.7.1
deskew using lines	HoughLinesP angle	LAB_6.ipynb → 5.7.2	LAB_1_3.ipynb 3.12.2
horizontal and vertical lines only	angle filter	LAB_6.ipynb → 5.7.3	
longest line, length, angle	max length	LAB_6.ipynb → 5.7.4	
erase or remove lines	draw over lines	LAB_6.ipynb → 5.7.7	
lines and circles together, shape finder	Hough both	LAB_5_6_LAB_TASKS.ipynb → X.4	LAB_6.ipynb 5.7.6

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
rectangles, screens, windows, panels, count, ON/OFF status, missing items versus expected	contours + approxPolyDP	LAB_5_6_LAB_TASKS.ipynb → 6.2 Detect Rectangular Objects, Count Them, Check Status / Missing Items	
triangle, square, circle, classify a shape by its corner count	approxPolyDP	LAB_6.ipynb → 7.2 Shape Detection and Counting	
extract an object using edges	edges + contours	LAB_6.ipynb → 7.3	

3G. Feature extraction, corners, SIFT matching (Lab 4, Lab 6)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
SIFT keypoints, how many, descriptors	SIFT	LAB_6.ipynb → 6.1, 6.2	
scale space, DoG pyramid / 26 neighbours, extrema / orientation / 128-D descriptor	SIFT theory in code	LAB_6.ipynb → 6.3 / 6.4 / 6.5 / 6.6	
match two images	descriptor matching	LAB_6.ipynb → 6.8 KNN Ratio Test (best default)	6.7 BFMatcher, 6.9 FLANN; LAB_4_5.ipynb 2.5.4

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
remove wrong matches, RANSAC, homography, locate an object in a scene	findHomography RANSAC	LAB_6.ipynb → 6.10	
recognise a known object from a reference, inventory, asset, bounding box	SIFT + RANSAC	LAB_5_6_LAB_TASKS.ipynb → 6.3 SIFT Feature Matching	LAB_6.ipynb 6.10
recognise an object in video or webcam	SIFT per frame	LAB_5_6_LAB_TASKS.ipynb → 6.4	
extract or rectify the object found in the scene	homography warp	LAB_6.ipynb → 6.11.1	
rotation angle or scale between two images	homography decomposition	LAB_6.ipynb → 6.11.2	
is SIFT invariant (test) / are these the same object / most similar image in a database	SIFT scoring	LAB_6.ipynb → 6.11.5 / 6.11.6 / 6.11.7	
keypoints only in a region / ORB alternative or SIFT unavailable / save matches	variants	LAB_6.ipynb → 6.11.8 / 6.11.9 / 6.11.10	LAB_4_5.ipynb 2.5.3 (ORB and SIFT keypoints)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
Harris corners / best N corners (Shi-Tomasi) / ORB and SIFT keypoints	corner detectors	LAB_4_5.ipynb → 2.5.1 / 2.5.2 / 2.5.3	
HOG, feature vector length	HOG	LAB_4_5.ipynb → 2.4.1 HOG with scikit-image	2.4.2 OpenCV only, 2.4.3 pedestrian detection
LBP, texture features	LBP	LAB_4_5.ipynb → 2.4.4	2.4.5 LBP per region (faces)
colour histogram, HSV histogram, feature vector	histogram features	LAB_4_5.ipynb → 2.4.6 BGR, 2.4.7 HSV	2.4.8 ROI histogram
compare two images by histogram	compareHist	LAB_4_5.ipynb → 2.4.9	LAB_1_3.ipynb 2.8.4
histogram of edge directions (HED) / intensity gradients (HIG) / texture energy and contrast / classify materials	handcrafted features	LAB_4_5.ipynb → 2.4.10 / 2.4.11 / 2.4.12 / 2.4.13	
compare two images using HOG, LBP and colour	feature comparison	LAB_4_5.ipynb → 4.4.1	

3H. Wavelets, video, signals (Lab 6)

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
DWT, approximation and detail, subbands, 1-level or 2-level	pywt dwt2	LAB_6.ipynb → 3.1, 3.2	LAB_5_6_LAB_TASKS.ipynb 6.11 2-D Wavelet Decomposition of an Image
reconstruct, inverse wavelet, LL only (blur), details only (edges)	idwt2	LAB_6.ipynb → 3.3	
wavelet denoise an image, PSNR	threshold details	LAB_6.ipynb → 3.4	
compress, keep the top X percent of coefficients	coefficient thresholding	LAB_6.ipynb → 3.5	
wavelet packet, all 16 subbands	WPT	LAB_6.ipynb → 3.6	
frequency over time, scalogram, CWT	cwt	LAB_6.ipynb → 3.7	
denoise audio or a 1-D signal, sensor anomaly, ECG, residuals	wavedec + threshold	LAB_5_6_LAB_TASKS.ipynb → 6.10 Wavelet Denoising and Anomaly Detection in 1-D Signals	LAB_6.ipynb 3.8 audio denoise
compare Haar vs db4 vs sym4 / wavelet edges / wavelet features / wavelet image fusion	variants	LAB_6.ipynb → 3.9 / 3.10 / 3.11 / 3.12	
video, frames, webcam	VideoCapture loop	by task: LAB_1_3_LAB_TASKS.ipynb 2.8 (enhancement), LAB_5_6_LAB_TASKS.ipynb 6.4 (SIFT), 6.6 (lanes), 6.8 (intruder)	LAB_6.ipynb 2.5 Video or Webcam Frame Processing

Question wording / keywords	Technique	PRIMARY: notebook → section	ALTERNATIVE / SPECIAL CASE
restricted zone, intruder, alarm, background subtraction, motion	absdiff / MOG2 + contours	LAB_5_6_LAB_TASKS.ipynb → 6.8 Restricted-Zone Intrusion Detection in Video	6.9 same idea with two still images

4. SIMILAR PROBLEMS: WHICH ONE SHOULD I USE?

4.1 "Separate the object from the background"

```

HOW IS THE OBJECT DIFFERENT FROM THE BACKGROUND?
|
|-- By COLOUR (red car, green leaf, yellow object)?
|   PRIMARY      LAB_5_6_LAB_TASKS.ipynb 5.4 (HSV)
|   ALTERNATIVE  LAB_4_5.ipynb 3.3.1;  unsure of the range: LAB_4_5.ipynb
3.3.2 (read HSV of a pixel)
|
|-- By BRIGHTNESS, lighting even?
|   |-- the question gives a number T    -> LAB_4_5.ipynb 3.2.1 (or
LAB_1_3.ipynb 1.11.1)
|   `-- no T given / two-hump histogram -> LAB_5_6_LAB_TASKS.ipynb 5.3
(Otsu)
|
|-- By BRIGHTNESS, but lighting UNEVEN (shadow, receipt, paper, text)?
|   PRIMARY  LAB_5_6_LAB_TASKS.ipynb 5.1 (compares global vs adaptive vs
Otsu)
|   tune block size and C: LAB_5_6_LAB_TASKS.ipynb 5.2
|   clean document binarization: LAB_4_5.ipynb 4.5.1
|
|-- Objects TOUCH each other?          -> LAB_5_6_LAB_TASKS.ipynb 5.7
watershed (tune with 5.8)
|-- Complex photo, many colours, "K clusters"? -> LAB_5_6_LAB_TASKS.ipynb
5.9 K-Means
|-- Start from a clicked point (seed)? -> LAB_5_6_LAB_TASKS.ipynb 5.6 region
growing
|-- Boundaries matter, edges visible?    -> LAB_4_5.ipynb 3.4.2 (edges to
filled mask); edge map: LAB_5_6_LAB_TASKS.ipynb 5.5

```

```
|-- Cut the foreground out with a rough rectangle? -> LAB_4_5.ipynb 3.8.1
GrabCut
`-- Method not named, image unknown? -> LAB_5_6_LAB_TASKS.ipynb 5.10 (three
methods compared) or LAB_4_5.ipynb 4.6.1
```

Do not use a plain global threshold (LAB_4_5.ipynb 3.2.1, LAB_1_3.ipynb 1.11.1) as first choice when the question mentions colour or uneven lighting. Use LAB_5_6_LAB_TASKS.ipynb 5.4 or 5.1 instead.

4.2 "Count / detect / find objects"

This mirrors the professor's own decision table in LAB_5_6_LAB_TASKS.ipynb X.5.

```
HOW MANY OBJECTS? WHAT DO THEY LOOK LIKE?
|
|-- Separate objects, brightness differs from the background
|   PRIMARY LAB_5_6_LAB_TASKS.ipynb X.1 (threshold + cleanup + contours;
also area, perimeter, centroid, crops)
|   simpler: LAB_1_3.ipynb 1.17.2      connected-components version:
LAB_4_5.ipynb 4.1.1
|-- Objects TOUCH and are roundish -> LAB_5_6_LAB_TASKS.ipynb 5.7 watershed
(LAB_4_5.ipynb 3.6.1 also counts)
|-- Objects are CIRCLES (coins, balls, wheels) -> LAB_5_6_LAB_TASKS.ipynb
6.7 Hough circles (radius groups)
|-- Objects differ by COLOUR -> LAB_5_6_LAB_TASKS.ipynb X.2
|-- Complex multi-colour photo -> LAB_5_6_LAB_TASKS.ipynb X.3 (K-Means,
then isolate a cluster)
|-- Lighting uneven -> adaptive threshold
(LAB_5_6_LAB_TASKS.ipynb 5.1) then X.1
|-- Straight-sided (screens, cards, windows), with ON/OFF or missing-item
check -> LAB_5_6_LAB_TASKS.ipynb 6.2
|-- Named shapes (triangle, square, circle) -> LAB_6.ipynb 7.2
`-- Known object shown in a reference image -> LAB_5_6_LAB_TASKS.ipynb 6.3
(SIFT)
```

Do not use LAB_5_6_LAB_TASKS.ipynb X.1 when objects touch. Touching objects merge into one contour and the count comes out too low. Use 5.7.

4.3 Lines, circles, rectangles, shapes

Situation	PRIMARY: notebook → section	Note
Straight lines, no extra context	LAB_5_6_LAB_TASKS.ipynb → 6.1	LAB_6.ipynb 5.2 HoughLines returns infinite lines (rho, theta); 5.3 HoughLinesP returns segments with end points
Road or lane lines	LAB_5_6_LAB_TASKS.ipynb → 6.6	adds Canny and a region of interest before Hough
Circles or coins	LAB_5_6_LAB_TASKS.ipynb → 6.7	too many circles: raise PARAM2; touching circles merged: raise MIN_DIST
Rectangles with a count or status	LAB_5_6_LAB_TASKS.ipynb → 6.2	if the question insists on Hough, draw lines with 6.1 first
Name the shape by its corner count	LAB_6.ipynb → 7.2	EPS controls approxPolyDP
"Implement Hough yourself"	LAB_6.ipynb → 5.4 lines, 5.6 circles	

4.4 Recognising, matching, aligning, stitching

TWO IMAGES OR A REFERENCE IMAGE INVOLVED?

```
|-- Is the known object somewhere in a scene (inventory, logo, product)?
|   -> LAB_5_6_LAB_TASKS.ipynb 6.3      (video or webcam: 6.4)
|-- Join overlapping photos into a panorama?
|   -> LAB_5_6_LAB_TASKS.ipynb 6.5      (just want it done in 6 lines:
6.5b)
|-- Align two views of the same scene?
|   -> LAB_1_3.ipynb 3.12.10 (ORB) or LAB_1_3_LAB_TASKS.ipynb 3.8
|-- Cut out or rectify the object found in the scene?
|   -> LAB_6.ipynb 6.11.1
|-- Many wrong matches?
|   -> LAB_6.ipynb 6.10 (RANSAC homography)
|-- "Are these the same object?" / rank database images
|   -> LAB_6.ipynb 6.11.6 / 6.11.7
`-- SIFT not installed or you want something faster?
    -> LAB_6.ipynb 6.11.9 (ORB alternative)
```

SIFT vs ORB: SIFT is the default for Lab 6 (`LAB_6.ipynb` section 6, `LAB_5_6_LAB_TASKS.ipynb` 6.3 to 6.5). ORB appears in `LAB_1_3.ipynb` 3.12.10, `LAB_1_3_LAB_TASKS.ipynb` 3.8 and `LAB_4_5.ipynb` 2.5.3, and is the fallback if `cv2.SIFT_create()` fails.

4.5 Bad brightness or contrast: which Lab 2 tool?

What the image looks like	Use	Notebook → section
Dark, detail hidden in shadows	gamma below 1, or log	<code>LAB_1_3.ipynb</code> 2.3.1 / 2.2.1; story version <code>LAB_1_3_LAB_TASKS.ipynb</code> 2.2
Washed out (too bright and flat)	gamma above 1, then stretch	<code>LAB_1_3.ipynb</code> 2.10.6
Narrow histogram, low contrast overall	histogram equalization	<code>LAB_1_3.ipynb</code> 2.9.1; story version <code>LAB_1_3_LAB_TASKS.ipynb</code> 2.1
Contrast uneven across the image, or equalization amplifies noise	CLAHE	<code>LAB_1_3.ipynb</code> 2.9.4
Colour image	equalize luminance only (Y of YCrCb, V of HSV, or L of LAB for CLAHE)	<code>LAB_1_3.ipynb</code> 2.9.3, 2.9.5
Linear spread of the existing range	contrast stretching (keeps the histogram shape)	<code>LAB_1_3.ipynb</code> 2.5.1
Make it look like a reference image	histogram matching	<code>LAB_1_3.ipynb</code> 2.9.6
Do not know	decision helper that prints a recommendation	<code>LAB_1_3.ipynb</code> 2.10.8

4.6 Geometry: what do you KNOW?

```
|-- A number (angle, shift, scale factor, shear)
|   shift LAB_1_3_LAB_TASKS.ipynb 3.4 | rotate 3.2 | scale 3.1 | shear 3.3
| several together 3.5
|-- A 2x2 matrix given          -> LAB_1_3.ipynb 3.7.1          a 3x3 matrix
given -> LAB_1_3.ipynb 3.8
|-- 3 point pairs              -> LAB_1_3_LAB_TASKS.ipynb 3.6 (affine, 6
unknowns)
|-- 4 point pairs / 4 corners -> LAB_1_3_LAB_TASKS.ipynb 3.7 (perspective)
|   corners NOT known         -> LAB_1_3.ipynb 3.12.4 (auto-detect)
`-- Two photos to join         -> LAB_5_6_LAB_TASKS.ipynb 6.5 or
LAB_1_3_LAB_TASKS.ipynb 3.8
```

"Straighten" can mean three different things. Rotated like a clock hand → rotation (LAB_1_3_LAB_TASKS.ipynb 3.2, use the negative angle). Rectangle looks like a trapezoid (far side smaller) → perspective (LAB_1_3_LAB_TASKS.ipynb 3.7). Leans like italic with parallel edges → shear (LAB_1_3_LAB_TASKS.ipynb 3.3).

Degrees of freedom (LAB_1_3_LAB_TASKS.ipynb , Part 5, section 5.4): translation 2, rigid 3, similarity 4, affine 6, projective 8. Points needed: affine 3, projective 4.

4.7 Noise removal

Noise or goal	Use	Notebook → section
Not sure which filter	compare them all	LAB_4_5.ipynb 2.2.5
Salt-and-pepper	median filter	LAB_1_3.ipynb 1.17.4, LAB_4_5.ipynb 2.2.5
Smooth Gaussian-type noise	Gaussian blur (bilateral if edges must stay sharp)	LAB_4_5.ipynb 2.2.2, LAB_1_3.ipynb 1.7.2
Noise before edge detection	blur first, then edges	LAB_6.ipynb 4.13.1
Wavelet denoising	threshold detail coefficients only, never the approximation	LAB_6.ipynb 3.4 (image), 3.8 or LAB_5_6_LAB_TASKS.ipynb 6.10 (1-D)

4.8 Edge detector

Question says	Use	Notebook → section
gradient, Gx, Gy, magnitude, direction	Sobel	<code>LAB_4_5.ipynb</code> 2.3.1, <code>LAB_6.ipynb</code> 4.1
thresholds, hysteresis, strong and weak edges	Canny	<code>LAB_5_6_LAB_TASKS.ipynb</code> 5.5
no thresholds given	Auto Canny	<code>LAB_4_5.ipynb</code> 2.3.7, <code>LAB_6.ipynb</code> 4.8
second derivative, zero crossing	LoG	<code>LAB_4_5.ipynb</code> 2.3.5, <code>LAB_6.ipynb</code> 4.10
compare detectors	side by side	<code>LAB_6.ipynb</code> 4.12

4.9 Document scanner: three versions

Version	Use when
PRIMARY <code>LAB_1_3.ipynb</code> 3.12.4	Corners must be found automatically (Canny + contour + approxPolyDP)
<code>LAB_1_3_LAB_TASKS.ipynb</code> 4.1 Document Scanner (Perspective Correction + Adaptive Threshold)	You type or pick the 4 corners and the result must be black-and-white
<code>LAB_6.ipynb</code> 7.1 Document Scanner	You want an automatic attempt with a MANUAL_POINTS fallback and a save path

5. MULTI-STEP / COMBINED QUESTIONS

First look for **one template that already does the whole pipeline**. Assemble pieces only if none fits.

Question pattern	BEST single template	If you must assemble
"Remove noise, segment, count, display"	<code>LAB_5_6_LAB_TASKS.ipynb</code> X.1 (blur, threshold, cleanup, contours, count)	add <code>LAB_4_5.ipynb</code> 2.2.5 or <code>LAB_1_3.ipynb</code> 1.17.4 first if the noise is salt-and-pepper
"Enhance contrast, then segment"	none	<code>LAB_1_3_LAB_TASKS.ipynb</code> 2.1, then <code>LAB_5_6_LAB_TASKS.ipynb</code> 5.1 or X.1
"Enhance, rotate, crop, annotate, save"	<code>LAB_1_3.ipynb</code> 4.0.1	
"Crop a region, equalize it, paste back"	<code>LAB_1_3.ipynb</code> 4.0.2	<code>LAB_1_3_LAB_TASKS.ipynb</code> 4.2
"Geometric plus photometric on one image, comparison grid"	<code>LAB_1_3.ipynb</code> 4.0.4	
"Show before/after histograms and statistics for any processing"	<code>LAB_1_3_LAB_TASKS.ipynb</code> 4.3	<code>LAB_1_3.ipynb</code> 4.0.3
"Scan a document and make it black-and-white"	<code>LAB_1_3_LAB_TASKS.ipynb</code> 4.1	<code>LAB_1_3.ipynb</code> 3.12.4, then <code>LAB_4_5.ipynb</code> 3.2.3
"Find the page, then straighten it"	<code>LAB_1_3.ipynb</code> 3.12.4 or <code>LAB_6.ipynb</code> 7.1	
"Segment by colour, then count"	<code>LAB_5_6_LAB_TASKS.ipynb</code> X.2	
"K-Means, isolate one region, count or extract"	<code>LAB_5_6_LAB_TASKS.ipynb</code> X.3	

Question pattern	BEST single template	If you must assemble
"Wavelet denoise, then Canny, then Hough"	LAB_6.ipynb 7.4	
"Denoise, then edges"	LAB_6.ipynb 4.13.1	
"Segment the largest object and crop it"	LAB_4_5.ipynb 4.2.2	
"Segment, then blur the background"	LAB_4_5.ipynb 4.3.1	
"Segment, then describe the object (area, perimeter, features)"	LAB_5_6_LAB_TASKS.ipynb X.1	LAB_4_5.ipynb 4.3.2
"Find the object in the scene, then rectify or crop it"	LAB_6.ipynb 6.11.1	
"Detect circles, then crop each"	LAB_6.ipynb 5.7.5	
"Lines and circles on the same image"	LAB_5_6_LAB_TASKS.ipynb X.4	LAB_6.ipynb 5.7.6
"Unknown multi-step question (A then B then C)"	LAB_4_5.ipynb 4.6.3 General Pipeline Skeleton	LAB_4_5.ipynb 4.6.1
"Do it for every image in a folder"	LAB_4_5.ipynb 4.6.2 (or LAB_1_3.ipynb 4.0.5) wrapped around the template	
"Intruder in a restricted zone, with alarm"	LAB_5_6_LAB_TASKS.ipynb 6.8	6.9 for two still images

6. DUPLICATE / OVERLAPPING TEMPLATE GUIDE

General rule: when the same technique appears in a *task* notebook

(LAB_1_3_LAB_TASKS.ipynb , LAB_5_6_LAB_TASKS.ipynb) and a *reference* notebook

(LAB_1_3.ipynb , LAB_4_5.ipynb , LAB_6.ipynb), use the task notebook if the question is a story with several parts, and the reference notebook if it names the exact variant.

Technique	Found in	PRIMARY	Use the others when
Global / Otsu / adaptive threshold	LAB_1_3.ipynb 1.11; LAB_1_3_LAB_TASKS.ipynb 1.7; LAB_4_5.ipynb 3.2; LAB_5_6_LAB_TASKS.ipynb 5.1 to 5.3 and X.1	Lab 5 style: LAB_5_6_LAB_TASKS.ipynb 5.1 or 5.3. Plain "threshold at T": LAB_4_5.ipynb 3.2.1	LAB_1_3.ipynb 1.11 or LAB_1_3_LAB_TASKS.ipynb 1.7 if the question is Lab 1 style (for example threshold then rotate)
Canny	LAB_1_3.ipynb 1.17.7; LAB_4_5.ipynb 2.3.6, 2.3.7, 3.4.1; LAB_6.ipynb 4.6 to 4.8; LAB_5_6_LAB_TASKS.ipynb 5.5	LAB_5_6_LAB_TASKS.ipynb 5.5 for threshold studies; LAB_4_5.ipynb 2.3.6 for plain	LAB_6.ipynb 4.6 if the manual-code version is wanted
Sobel / Prewitt / Scharr / Laplacian	LAB_4_5.ipynb 2.3; LAB_6.ipynb 4.1 to 4.5	LAB_4_5.ipynb 2.3 for plain use	LAB_6.ipynb section 4 for hand calculation, magnitude and direction, binary edge map, per-channel, ROI, overlay
Histogram equalization	LAB_1_3.ipynb 1.14 and 2.9; LAB_1_3_LAB_TASKS.ipynb 2.1	LAB_1_3_LAB_TASKS.ipynb 2.1 for story questions (X-ray); LAB_1_3.ipynb 2.9.x for a named variant	LAB_1_3.ipynb 1.14 only if the question is Lab 1 flavoured
Colour histogram	LAB_1_3.ipynb 2.8.2; LAB_4_5.ipynb 2.4.6, 2.4.7, 2.4.9	LAB_1_3.ipynb 2.8.2 to display; LAB_4_5.ipynb 2.4.6 / 2.4.7 as a feature vector; 2.4.9 to compare	
Blur / smoothing	LAB_1_3.ipynb 1.7; LAB_4_5.ipynb 2.2	LAB_1_3.ipynb 1.7.1 Lab 1 style; LAB_4_5.ipynb 2.2.x Lab 4 style (box vs Gaussian, convolution)	
Rotation	LAB_1_3.ipynb 1.12 and 3.3; LAB_1_3_LAB_TASKS.ipynb 3.2	LAB_1_3_LAB_TASKS.ipynb 3.2	LAB_1_3.ipynb 1.12.1 only for a simple centre rotation

Technique	Found in	PRIMARY	Use the others when
Resize / scale	LAB_1_3.ipynb 1.6 and 3.4; LAB_1_3_LAB_TASKS.ipynb 3.1	LAB_1_3.ipynb 1.6 for resize to a size or aspect ratio; LAB_1_3_LAB_TASKS.ipynb 3.1 for a scale matrix or keep centred	
Document scanner	LAB_1_3.ipynb 3.12.4; LAB_1_3_LAB_TASKS.ipynb 4.1; LAB_6.ipynb 7.1	LAB_1_3.ipynb 3.12.4	see guide 4.9
Counting objects via contours	LAB_1_3.ipynb 1.17.2; LAB_4_5.ipynb 3.4.3, 4.1.1; LAB_6.ipynb 4.13.5, 7.2; LAB_5_6_LAB_TASKS.ipynb X.1, 6.2	LAB_5_6_LAB_TASKS.ipynb X.1	see guide 4.2
Morphology (erode, dilate, open, close)	LAB_4_5.ipynb 3.9.1 (the only dedicated cell)	LAB_4_5.ipynb 3.9.1	morphology is also used inside LAB_5_6_LAB_TASKS.ipynb 5.7, X.1, 5.4; LAB_6.ipynb 4.13.6 for boundary extraction
Watershed	LAB_4_5.ipynb 3.6.1; LAB_5_6_LAB_TASKS.ipynb 5.7, 5.8	LAB_5_6_LAB_TASKS.ipynb 5.7	5.8 for the tuning table
K-Means	LAB_4_5.ipynb 3.7.x; LAB_5_6_LAB_TASKS.ipynb 5.9, X.3	LAB_5_6_LAB_TASKS.ipynb 5.9	LAB_4_5.ipynb 3.7.4 to choose K
HSV colour mask	LAB_1_3.ipynb 1.17.1; LAB_1_3_LAB_TASKS.ipynb 2.6; LAB_4_5.ipynb 3.3.1; LAB_5_6_LAB_TASKS.ipynb 5.4, X.2	LAB_5_6_LAB_TASKS.ipynb 5.4	X.2 to count
SIFT / ORB	LAB_4_5.ipynb 2.5.3, 2.5.4; LAB_6.ipynb section 6; LAB_5_6_LAB_TASKS.ipynb 6.3 to 6.5	LAB_6.ipynb section 6 for theory or variants; LAB_5_6_LAB_TASKS.ipynb 6.3 for recognition	LAB_4_5.ipynb 2.5.x for a quick keypoint demo
Panorama / homography	LAB_1_3.ipynb 3.12.10; LAB_1_3_LAB_TASKS.ipynb 3.8; LAB_6.ipynb 6.11.3, 6.11.4; LAB_5_6_LAB_TASKS.ipynb 6.5	LAB_5_6_LAB_TASKS.ipynb 6.5	LAB_1_3_LAB_TASKS.ipynb 3.8 for typed-in points

Technique	Found in	PRIMARY	Use the others when
Hough lines	LAB_6.ipynb 5.2, 5.3; LAB_5_6_LAB_TASKS.ipynb 6.1, 6.6	LAB_5_6_LAB_TASKS.ipynb 6.1	6.6 for lanes; LAB_6.ipynb 5.2 / 5.3 for variants
Deskew	LAB_1_3.ipynb 3.12.2; LAB_6.ipynb 5.7.2	LAB_1_3.ipynb 3.12.2	LAB_6.ipynb 5.7.2 if the question says Hough
Wavelets	LAB_6.ipynb section 3; LAB_5_6_LAB_TASKS.ipynb 6.10, 6.11	LAB_6.ipynb section 3	LAB_5_6_LAB_TASKS.ipynb 6.10 for 1-D sensor data
Video	LAB_1_3_LAB_TASKS.ipynb 2.8; LAB_5_6_LAB_TASKS.ipynb 6.4, 6.6, 6.8; LAB_6.ipynb 2.5	by task, see table 3H	
Image statistics	LAB_1_3.ipynb 1.10; LAB_1_3_LAB_TASKS.ipynb 1.9	LAB_1_3_LAB_TASKS.ipynb 1.9	

7. WHAT TO CHANGE IN THE MOST LIKELY TEMPLATES

After you copy a cell, edit only these. Variable names are the real ones in the notebooks. Every cell also needs `IMAGE_PATH` (or the equivalent path variable) set first.

Template (notebook → section)	Use when	Do NOT use if	Change these values	If the output looks wrong
LAB_5_6_LAB_TASKS.ipynb → 5.1 Global vs Adaptive vs Otsu	text or objects on a background with uneven lighting; "compare thresholds"	the question is about colour	GLOBAL_THRESHOLDS, FOREGROUND_IS_DARK, BLOCK_SIZE, C	mask inverted: flip FOREGROUND_IS_DARK; noisy: raise C or blur first
LAB_5_6_LAB_TASKS.ipynb → 5.3 Otsu	threshold must be chosen automatically; two-hump histogram	lighting is uneven	OBJECTS_ARE_DARK, BLUR_KSIZE, the preprocessing variants dictionary	all white or all black: flip OBJECTS_ARE_DARK
LAB_5_6_LAB_TASKS.ipynb → 5.4 HSV colour segmentation	the object is identified by colour	colour is not the distinguishing feature	LOWER_FINAL / UPPER_FINAL, SAMPLE_PIXEL = (x, y), TOL, WRAP_HUE = True for red (set LOWER2 / UPPER2)	parts missing: widen the HSV range; red cut in half: WRAP_HUE

Template (notebook → section)	Use when	Do NOT use if	Change these values	If the output looks wrong
LAB_5_6_LAB_TASKS.ipynb → 5.5 Canny hysteresis study	edge map, boundary, "change the thresholds"	you need closed regions (use LAB_4_5.ipynb 3.4.2)	THRESHOLD_PAIRS, BLUR_KSIZE, USE_AUTO	too many edges: raise thresholds; too few: lower them
LAB_5_6_LAB_TASKS.ipynb → 5.6 Region growing	start from a seed, contiguous similar region	no seed is given and none can be chosen	SEEDS, THRESHOLDS, CONNECTIVITY, COMPARE_TO	leaks everywhere: lower threshold
LAB_5_6_LAB_TASKS.ipynb → 5.7 Watershed	roundish objects that touch	objects are separate (use X.1)	OBJECTS_ARE_DARK, OPEN_ITER, DILATE_ITER, DIST_RATIO, KERNEL_SIZE	objects merged: raise DIST_RATIO; over-split: lower DIST_RATIO
LAB_5_6_LAB_TASKS.ipynb → 5.9 K-Means	clustering, reduce colours, K = 2, 4, 6	you need exact object boundaries by brightness	K_VALUES, ATTEMPTS, MAX_WIDTH, MAX_ITER, EPSILON	
LAB_5_6_LAB_TASKS.ipynb → X.1 Detect, count, measure objects	separate objects, count, area, perimeter, centroid, crops	objects touch (use 5.7) or are circles (use 6.7)	THRESH_METHOD (otsu / adaptive / fixed), OBJECTS_ARE_DARK, MIN_AREA, MAX_AREA, OPEN_ITER, CLOSE_ITER, SAVE_CROPS	noise counted as objects: raise MIN_AREA or OPEN_ITER; mask inverted: flip OBJECTS_ARE_DARK
LAB_5_6_LAB_TASKS.ipynb → X.2 Colour segmentation then count	coloured objects to count or locate		COLOR_RANGES, MIN_AREA, KERNEL_SIZE	
LAB_5_6_LAB_TASKS.ipynb → X.3 K-Means then isolate	isolate one colour cluster, count or extract		K, TARGET_RGB, CLUSTER_ID, MIN_AREA	
LAB_5_6_LAB_TASKS.ipynb → 6.1 Hough lines	straight boundaries, horizontal or vertical classification		MODE, CANNY_LO / CANNY_HI, HOUGH_THRESH, MIN_LINE_LEN, MAX_LINE_GAP, ANGLE_TOL, RHO, THETA	too many lines: raise HOUGH_THRESH; too few: lower it
LAB_5_6_LAB_TASKS.ipynb → 6.2 Rectangles, count, status	screens, windows, panels; count versus expected	shapes are not four-sided	EXPECTED_COUNT, MIN_AREA, MAX_AREA, ASPECT_MIN / ASPECT_MAX, ON_BRIGHTNESS, CANNY_LO / CANNY_HI, APPROX_EPS (0.02 typical), ROW_TOL	
LAB_5_6_LAB_TASKS.ipynb → 6.3 SIFT recognition	known object from a reference image	no reference image exists	SCENE_PATHS, REFERENCES, RATIO, MIN_INLIERS, RANSAC_THRESH, SHOW_MATCHES	object not found: lower MIN_INLIERS, RATIO 0.8, check reference quality
LAB_5_6_LAB_TASKS.ipynb → 6.5 Panorama	overlapping photos	images do not overlap	IMAGE_PATHS, RESIZE_WIDTH, RATIO, RANSAC_THRESH, CROP_BLACK	fallback: 6.5b built-in Stitcher

Template (notebook → section)	Use when	Do NOT use if	Change these values	If the output looks wrong
LAB_5_6_LAB_TASKS.ipynb → 6.6 Lane detection	road or lane lines, image or video		IMAGE_PATHS or VIDEO_PATH + MODE, ROI_FRAC, CANNY_LO / CANNY_HI, HOUGH_THRESH, MIN_LINE_LEN, MAX_LINE_GAP, MIN_SLOPE, Y_TOP_FRAC	
LAB_5_6_LAB_TASKS.ipynb → 6.7 Hough circles	circular objects, count, radius groups		MEDIAN_BLUR, DP, MIN_DIST, PARAM1, PARAM2, MIN_RADIUS, MAX_RADIUS, GAP_TOL	too many circles: raise PARAM2; merged circles: raise MIN_DIST
LAB_5_6_LAB_TASKS.ipynb → 6.8 Intrusion in a zone	video, restricted area, alarm		VIDEO_PATH (a file, or 0 for webcam), ZONE_FRAC, METHOD, MIN_AREA, MIN_OVERLAP, WARMUP_FRAMES, DIFF_THRESH, OUTPUT_VIDEO	
LAB_5_6_LAB_TASKS.ipynb → 6.10 Wavelet 1-D signal	sensor data, denoise, anomalies	it is an image (use 6.11)	SENSOR_FILE, WAVELET, LEVEL, THRESH_MODE, K_SIGMA, THRESH_SCALE	
LAB_1_3_LAB_TASKS.ipynb → 2.1 Histogram + equalization	low-contrast or underexposed image		METHOD (global or clahe), IS_COLOR (True = equalize brightness only), CLIP_LIMIT, TILE_GRID	colours shifted: set IS_COLOR = True
LAB_1_3_LAB_TASKS.ipynb → 2.2 Log + gamma	reveal dark detail, brighten or darken		GAMMA (below 1 brightens, above 1 darkens), AS_GRAY, GAMMAS list	
LAB_1_3_LAB_TASKS.ipynb → 3.2 Rotation without cropping	rotate with no cut corners, un-tilt		ANGLE_DEG (positive = counter-clockwise; to undo a +45 degree tilt use -45), EXPAND	wrong direction: flip the sign of the angle
LAB_1_3_LAB_TASKS.ipynb → 3.6 Affine from 3 point pairs	3 landmarks given	4 points given (use 3.7)	SRC_PTS (in the distorted image), DST_PTS (where they should be), OUT_SIZE	
LAB_1_3_LAB_TASKS.ipynb → 3.7 Perspective correction	4 corners of a flat object		SRC_PTS (any order, the code orders them), OUT_W / OUT_H (None = automatic), FORCE_SQUARE	twisted output: point order must be top-left, top-right, bottom-right, bottom-left; SRC and DST not swapped
LAB_1_3_LAB_TASKS.ipynb → 3.8 Panorama (typed points or ORB)	two photos, homography		IMG1_PATH (reference), IMG2_PATH (warped), PTS1, PTS2 (same physical points, same order), USE_AUTO_POINTS, BLEND_OVERLAP	black canvas: PTS1 and PTS2 swapped

Template (notebook → section)	Use when	Do NOT use if	Change these values	If the output looks wrong
<code>LAB_1_3_LAB_TASKS.ipynb</code> → 4.1 Document scanner	4 corners known, black-and-white result	corners must be auto-detected (use <code>LAB_1_3.ipynb</code> 3.12.4)	SRC_PTS, BLOCK_SIZE, C, PRE_BLUR	
<code>LAB_1_3.ipynb</code> → 3.12.4 Auto document scanner	corners must be found automatically		MIN_AREA_FRAC	no page found: lower MIN_AREA_FRAC
<code>LAB_1_3.ipynb</code> → 1.17.2 Threshold, boxes, count	simple count with bounding boxes		MIN_AREA	
<code>LAB_4_5.ipynb</code> → 4.1.1 Count with connected components	count and label regions		INVERT, MIN_AREA, KSIZE	
<code>LAB_6.ipynb</code> → 7.2 Shape detection and counting	triangle, square, circle by corner count		MIN_AREA, EPS	
<code>LAB_6.ipynb</code> → 7.1 Document scanner	automatic with manual fallback		MANUAL_POINTS (four page corners), OUT_PATH	

Generic fixes (`LAB_1_3_LAB_TASKS.ipynb` section 5.3 and `LAB_5_6_LAB_TASKS.ipynb` section 4.3): wrong colours → BGR vs RGB; kernel error → kernel size must be odd; mask inverted → flip THRESH_BINARY / THRESH_BINARY_INV; too much noise → blur, raise C, more OPEN_ITER, raise MIN_AREA; missing parts → lower C or threshold, add CLOSE_ITER, widen HSV range.

8. NOTEBOOK-BY-NOTEBOOK MAP

`LAB_1_3.ipynb` (Labs 1-3 Complete Exam Code Reference)

Section	Contents
0	Scenario to code quick-reference table (clickable)
1.1 to 1.5	Setup · display · properties, pixels, channels · grayscale · colour spaces
1.6 to 1.9	Resizing · filtering and sharpening · cropping and ROI · text and shapes
1.10 to 1.15	Pandas / NumPy statistics · thresholding · basic rotation · blending and arithmetic · histogram equalization (Lab 1 version) · saving and folders

Section	Contents
1.16	Lab 1 theory cheat sheet (likely MCQ material)
1.17.1 to 1.17.9	Lab 1 exam scenarios: colour mask, count with boxes, preprocessing pipeline, salt-and-pepper add and remove, crop + resize + label, mask a region, Canny, grid comparison, pixel counting
2.1 to 2.4	Negative · log · gamma · transfer curves
2.5 to 2.7	Contrast stretching and brightness · piecewise linear · intensity-level and bit-plane slicing
2.8, 2.9	Histogram compute, plot, statistics · histogram equalization, CLAHE, matching
2.10.1 to 2.10.9	Lab 2 exam scenarios (auto-fix, compare all, ROI, conditional, stats, washed-out, X-ray, decision helper, one channel)
3.1 to 3.6	Pixel coordinates · translation · rotation · scaling · reflection · shearing
3.7 to 3.11	Any 2x2 matrix · homogeneous coordinates and composition · rigid and similarity · affine from 3 pairs · perspective
3.12.1 to 3.12.11	Lab 3 exam scenarios (straighten, auto-deskew, perspective, document scanner, region only, remove black corners, collage, point transform, billboard, ORB alignment, printed matrices)
4.0.1 to 4.0.6	Combined scenarios
5.1, 5.2, 5.3	Function signatures · pitfalls that cost marks · likely MCQ facts (Labs 2 and 3)

LAB_1_3_LAB_TASKS.ipynb (Labs 1-3 task templates)

Section	Contents
1.0 to 1.9	Setup · class manager · nested dictionary · safe loading · canvas drawing · blur + centre ROI · transparent overlay · global vs adaptive threshold + rotate · bitwise compositing · pandas profiling
2.1 to 2.8	Histogram + equalization · log + gamma · piecewise / stretching / negative / slicing · heatmap · colour balance · threshold or mask dense regions · two-image fusion · video enhancement
3.1 to 3.10	Scaling · rotation without cropping · shear · translation · combine into one matrix · affine from 3 points · perspective correction · panorama · full hierarchy into a frame · warp without warpAffine
4.1 to 4.4	Document scanner · enhance a selected region · multi-step pipeline with histograms and stats · finding pixel coordinates
Part 5 (5.1 to 5.5)	"If the question says" lookup · technique to template · common parameter changes · concept and MCQ notes · frequent mistakes

LAB_4_5.ipynb (Labs 4-5 Complete Exam Code Reference)

Section	Contents
1.2	Scenario to code quick-reference table (clickable)
2.2.1 to 2.2.5	Lab 4 filtering: box blur, Gaussian, kernel gallery, convolution types, noise removal comparison
2.3.1 to 2.3.7	Lab 4 edges: Sobel, Scharr, Prewitt, Laplacian, LoG, Canny, Auto Canny
2.4.1 to 2.4.13	Lab 4 features: HOG (3), LBP (2), colour histograms (3), compare by histogram, HED, HIG, texture energy, material classification

Section	Contents
2.5.1 to 2.5.4	Corners and keypoints: Harris, Shi-Tomasi, ORB / SIFT, matching
3.2.1 to 3.2.3	Lab 5 thresholding: global, Otsu, adaptive
3.3, 3.4, 3.5	Colour segmentation (3.3.1 to 3.3.3) · edge-based (3.4.1 to 3.4.3) · region growing (3.5.1, 3.5.2)
3.6, 3.7, 3.8, 3.9	Watershed · clustering (K-Means variants, elbow, mean shift) · GrabCut, superpixels, snakes, DeepLab · morphology
4.1 to 4.6	Combined scenarios: counting, extract and replace, region-specific operations, compare and analyse, uneven-lighting documents, templates for unknown questions (4.6.1 every method, 4.6.2 folder, 4.6.3 pipeline skeleton)
5.1 to 5.4	What to change to get this effect · manual code bugs and fixes · MCQ facts · troubleshooting errors

LAB_6.ipynb (Lab 6 Complete Exam Code Reference)

Section	Contents
1	Exam wording to section quick-reference table
2.1 to 2.5	Setup, practice images, display helper, type and colour conversions, video or webcam frame processing
3.1 to 3.13	Wavelets: single and multi-level DWT, inverse, denoise, compress, packet, CWT, audio denoise, families, edges, features, fusion, exam-scenario table
4.1 to 4.13	Boundary detection: Sobel (4 cells), Prewitt / Roberts / Scharr, Canny (3 cells), LoG (3 cells), comparison, 4.13.1 to 4.13.10 scenarios

Section	Contents
5.1 to 5.7	Hough: concept, HoughLines, HoughLinesP, from scratch (lines, circles), Hough circles, 5.7.1 to 5.7.7 scenarios
6.1 to 6.11	SIFT: keypoints, parameters, scale space, extrema, orientation, descriptor, matching (BF, KNN, FLANN), RANSAC, 6.11.1 to 6.11.10 scenarios
7.1 to 7.4	Combined: document scanner, shape detection and counting, extract object using edges, wavelet denoise then Canny then Hough
8.1 to 8.4	Key concepts · function cheat sheet · parameter effects · exam traps

LAB_5_6_LAB_TASKS.ipynb (Labs 5-6 task templates)

Section	Contents
5.1 to 5.10	Threshold comparison · adaptive parameter grid · Otsu · HSV colour · Canny study · region growing · watershed · watershed tuning · K-Means · compare three methods
6.1 to 6.11	Hough lines · rectangles and status · SIFT recognition · SIFT in video · panorama (6.5, 6.5b) · lanes · Hough circles · zone intrusion in video · zone check with two stills (6.9) · 1-D wavelet · 2-D wavelet
X.1 to X.5	Count / measure / extract · colour then count · K-Means then isolate · Hough lines + circles · decision guide for counting
Part 4 (4.1 to 4.6)	"If the question says" lookup · technique to OpenCV function · parameter changes · theory quick-answers · exam-day workflow · before-the-exam checklist

9. THEORY / MCQ: WHERE THE ANSWERS ARE

Topic	Notebook → section
Lab 1 definitions, libraries, DIP steps	<code>LAB_1_3.ipynb</code> → 1.16 Lab 1 Theory Cheat Sheet
Lab 2 and Lab 3 MCQ facts, function signatures, pitfalls	<code>LAB_1_3.ipynb</code> → 5.1 Function signatures, 5.2 Pitfalls that cost marks, 5.3 Likely MCQ facts
Point vs neighbourhood transforms, log vs gamma, equalization vs stretching vs CLAHE, global vs adaptive vs Otsu, DoF and what each transform preserves, inverse mapping	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 5.4 Concept / MCQ notes
Transformation hierarchy (scale, rigid, similarity, affine, projective)	<code>LAB_1_3_LAB_TASKS.ipynb</code> → the table under "The transformation hierarchy (know this for MCQs)" at the start of Part 3, and 5.4
Mistakes to check first when output looks wrong (Labs 1-3)	<code>LAB_1_3_LAB_TASKS.ipynb</code> → 5.5 Frequent mistakes
Feature extraction and segmentation overviews	<code>LAB_4_5.ipynb</code> → 2.1 Feature Extraction Overview, 3.1 Segmentation Overview
Labs 4-5 MCQ facts, parameter effects, bugs, errors	<code>LAB_4_5.ipynb</code> → 5.1 What to Change to Get This Effect, 5.2 Manual Code Bugs and Fixes, 5.3 MCQ Facts, 5.4 Troubleshooting Errors
Lab 6 concepts, function list, parameter effects, traps	<code>LAB_6.ipynb</code> → 8.1 Key Concepts, 8.2 Function Cheat Sheet, 8.3 Parameter Effects, 8.4 Exam Traps
Labs 5-6 theory quick-answers (why global fails, Otsu, HSV, Canny steps, region growing, watershed stages, K-Means, Hough, SIFT, homography, wavelet vs Fourier, background subtraction)	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 4.4 Theory quick-answers
Written "explain / justify" parts for any Lab 5 or Lab 6 task	<code>LAB_5_6_LAB_TASKS.ipynb</code> → the "Written-answer cheat lines" under each template (5.1 to 6.11, X.1 to X.4)

Topic	Notebook → section
Symptom → parameter to try first	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 4.3 Common parameter changes; <code>LAB_1_3_LAB_TASKS.ipynb</code> → 5.3
Exam-day workflow	<code>LAB_5_6_LAB_TASKS.ipynb</code> → 4.5

Ten traps that cost marks (from `LAB_1_3_LAB_TASKS.ipynb` 5.5 and `LAB_5_6_LAB_TASKS.ipynb` 4.3):

1. Size arguments (`dsize`, `resize`) are (`width`, `height`), but `img.shape` is (`height`, `width`).
2. Points are (`x`, `y`); NumPy indexing is `[row, col] = [y, x]`.
3. `uint8` overflow: `img + 50` wraps around. Use `cv2.add`, or float then `np.clip`.
4. `cv2.imread` returns `None` silently on a bad path.
5. Matplotlib shows grayscale in colour unless `cmap="gray"`.
6. Kernel size and adaptive block size must be odd.
7. OpenCV is BGR; Matplotlib is RGB. HSV hue range in OpenCV is 0 to 179.
8. Perspective corner order is top-left, top-right, bottom-right, bottom-left; do not swap source and destination.
9. Mask looks inverted: flip `THRESH_BINARY` and `THRESH_BINARY_INV`.
10. Rotation angle: positive = counter-clockwise (`getRotationMatrix2D`). Wrong direction: flip the sign.

10. GAPS AND WHAT TO DO IN YOUR 10-MINUTE INTERNET WINDOW

Not found in any of the five notebooks (checked by searching the code): template matching (`cv2.matchTemplate`), Haar cascade / face detection (`CascadeClassifier`), convex hull, `minEnclosingCircle`, `fitLine`. If the syllabus breakdown your teacher posts lists any of these, download an OpenCV tutorial for it.

Lab 2 manual: it was not provided, but Lab 2 topics (log, gamma, piecewise, histogram) are covered in `LAB_1_3.ipynb` section 2 and `LAB_1_3_LAB_TASKS.ipynb` 2.1 to 2.8.

Libraries to check before the exam (run `import cv2; print(cv2.__version__); cv2.SIFT_create()` and `import pywt`, as `LAB_5_6_LAB_TASKS.ipynb` 4.6 advises):

Library	Needed by
<code>pywt</code> (PyWavelets)	<code>LAB_6.ipynb</code> 3.1 to 3.12 and 7.4; <code>LAB_5_6_LAB_TASKS.ipynb</code> 6.10, 6.11 (6.10 has a Haar fallback if <code>pywt</code> is missing)
<code>skimage</code> (scikit-image)	<code>LAB_4_5.ipynb</code> 2.4.1, 2.4.4, 2.4.5, 2.4.13, 3.8.2, 3.8.3, 4.3.2, 4.4.1; <code>LAB_1_3.ipynb</code> 1.2.5
<code>sklearn</code> (scikit-learn)	<code>LAB_4_5.ipynb</code> 2.4.13
<code>torch</code>	<code>LAB_4_5.ipynb</code> 3.8.4 only (DeepLab); skip
<code>scipy</code>	<code>LAB_6.ipynb</code> 3.8, 6.4
<code>pandas</code>	<code>LAB_1_3_LAB_TASKS.ipynb</code> 1.0, 1.9, 4.3; <code>LAB_1_3.ipynb</code> 1.10.1

Sample inputs the templates expect (from `LAB_5_6_LAB_TASKS.ipynb` 4.6): touching coins, an unevenly lit document, a yellow car, a brain MRI, a wildlife photo, a road / lane photo, a panorama image sequence, a small video.

11. EMERGENCY EXAM CHEAT SHEET

Format: **question says** → **notebook + section**. Where two are shown, the first is the primary.

BASICS (LAB 1)

```
load / show / fix colours ..... LAB_1_3_LAB_TASKS.ipynb 1.3
size / channels / pixel ..... LAB_1_3.ipynb 1.3.1
split channels ..... LAB_1_3.ipynb 1.3.2
grayscale ..... LAB_1_3.ipynb 1.4.1
HSV / LAB / YCrCb ..... LAB_1_3.ipynb 1.5.1
resize ..... LAB_1_3.ipynb 1.6
blur / sharpen ..... LAB_1_3.ipynb 1.7
crop / ROI ..... LAB_1_3.ipynb 1.8
text / shapes ..... LAB_1_3.ipynb 1.9
draw on blank canvas ..... LAB_1_3_LAB_TASKS.ipynb 1.4
transparent caption ..... LAB_1_3_LAB_TASKS.ipynb 1.6
pandas stats ..... LAB_1_3_LAB_TASKS.ipynb 1.9
bitwise cut and paste ..... LAB_1_3_LAB_TASKS.ipynb 1.8
blend two images ..... LAB_1_3.ipynb 1.13.2
```

folder of images LAB_1_3.ipynb 1.15.2
 class / dictionary LAB_1_3_LAB_TASKS.ipynb 1.1 / 1.2

POINT TRANSFORMS (LAB 2)

negative LAB_1_3.ipynb 2.1.1
 log LAB_1_3.ipynb 2.2
 gamma LAB_1_3.ipynb 2.3
 log + gamma story LAB_1_3_LAB_TASKS.ipynb 2.2
 contrast stretching LAB_1_3.ipynb 2.5
 piecewise linear LAB_1_3.ipynb 2.6
 intensity slicing / bit planes ... LAB_1_3.ipynb 2.7
 histogram LAB_1_3.ipynb 2.8
 equalization LAB_1_3.ipynb 2.9 (story:
 LAB_1_3_LAB_TASKS.ipynb 2.1)
 CLAHE LAB_1_3.ipynb 2.9.4
 heatmap LAB_1_3_LAB_TASKS.ipynb 2.4
 colour cast LAB_1_3_LAB_TASKS.ipynb 2.5
 mask brightest / colour range LAB_1_3_LAB_TASKS.ipynb 2.6
 fuse two images LAB_1_3_LAB_TASKS.ipynb 2.7
 video enhancement LAB_1_3_LAB_TASKS.ipynb 2.8
 enhance a region only LAB_1_3.ipynb 2.10.3
 before / after stats LAB_1_3_LAB_TASKS.ipynb 4.3

GEOMETRY (LAB 3)

shift / translate LAB_1_3_LAB_TASKS.ipynb 3.4
 rotate (no cropping) LAB_1_3_LAB_TASKS.ipynb 3.2
 scale / zoom LAB_1_3_LAB_TASKS.ipynb 3.1
 flip / mirror LAB_1_3.ipynb 3.5
 shear LAB_1_3_LAB_TASKS.ipynb 3.3
 given 2x2 matrix LAB_1_3.ipynb 3.7.1
 compose / master matrix LAB_1_3_LAB_TASKS.ipynb 3.5
 rigid / similarity LAB_1_3_LAB_TASKS.ipynb 3.5
 3 point pairs (affine) LAB_1_3_LAB_TASKS.ipynb 3.6
 4 corners (perspective) LAB_1_3_LAB_TASKS.ipynb 3.7
 pick points, no numbers LAB_1_3_LAB_TASKS.ipynb 4.4
 scan document (auto corners) LAB_1_3.ipynb 3.12.4
 scan document (typed corners) LAB_1_3_LAB_TASKS.ipynb 4.1
 deskew LAB_1_3.ipynb 3.12.2
 place image in a frame LAB_1_3_LAB_TASKS.ipynb 3.9
 warp without warpAffine LAB_1_3_LAB_TASKS.ipynb 3.10
 where does a point go LAB_1_3.ipynb 3.12.8

FILTERS, EDGES, FEATURES (LAB 4)

box / Gaussian blur	LAB_4_5.ipynb 2.2.1 / 2.2.2	
kernel / sharpen / emboss	LAB_4_5.ipynb 2.2.3	
convolution output size	LAB_4_5.ipynb 2.2.4	
remove noise	LAB_4_5.ipynb 2.2.5	
Sobel	LAB_4_5.ipynb 2.3.1	(manual / by hand:
LAB_6.ipynb 4.1 / 4.4)		
Scharr / Prewitt	LAB_4_5.ipynb 2.3.2 / 2.3.3	
Laplacian / LoG	LAB_4_5.ipynb 2.3.4 / 2.3.5	
Canny	LAB_5_6_LAB_TASKS.ipynb 5.5	(plain:
LAB_4_5.ipynb 2.3.6)		
Canny, no thresholds given	LAB_4_5.ipynb 2.3.7	
compare edge detectors	LAB_6.ipynb 4.12	
HOG	LAB_4_5.ipynb 2.4.1	
LBP	LAB_4_5.ipynb 2.4.4	
colour histogram feature	LAB_4_5.ipynb 2.4.6 / 2.4.7	
compare images by histogram	LAB_4_5.ipynb 2.4.9	
compare images HOG + LBP + colour	LAB_4_5.ipynb 4.4.1	
Harris / Shi-Tomasi	LAB_4_5.ipynb 2.5.1 / 2.5.2	
SEGMENTATION (LAB 5)		
threshold at a value	LAB_4_5.ipynb 3.2.1	
Otsu	LAB_5_6_LAB_TASKS.ipynb 5.3	
uneven light / adaptive	LAB_5_6_LAB_TASKS.ipynb 5.1	(grid: 5.2)
segment by colour	LAB_5_6_LAB_TASKS.ipynb 5.4	
seed / region growing	LAB_5_6_LAB_TASKS.ipynb 5.6	
touching objects / watershed	LAB_5_6_LAB_TASKS.ipynb 5.7	(tuning: 5.8)
K-Means	LAB_5_6_LAB_TASKS.ipynb 5.9	
remove background	LAB_4_5.ipynb 3.8.1	
clean mask / morphology	LAB_4_5.ipynb 3.9.1	
unknown image, choose a method ...	LAB_5_6_LAB_TASKS.ipynb 5.10	
count objects	LAB_5_6_LAB_TASKS.ipynb X.1	(see guide
4.2)		
measure area / perimeter	LAB_5_6_LAB_TASKS.ipynb X.1	
colour then count	LAB_5_6_LAB_TASKS.ipynb X.2	
crop the largest object	LAB_4_5.ipynb 4.2.2	
unknown multi-step pipeline	LAB_4_5.ipynb 4.6.3	
LINES, CIRCLES, SHAPES, MATCHING (LAB 6)		
straight lines	LAB_5_6_LAB_TASKS.ipynb 6.1	
lanes	LAB_5_6_LAB_TASKS.ipynb 6.6	
circles / coins	LAB_5_6_LAB_TASKS.ipynb 6.7	
rectangles / screens	LAB_5_6_LAB_TASKS.ipynb 6.2	
triangle / square / circle	LAB_6.ipynb 7.2	

Hough from scratch LAB_6.ipynb 5.4 / 5.6
 SIFT keypoints LAB_6.ipynb 6.1 / 6.2
 match two images LAB_6.ipynb 6.8
 RANSAC / locate in scene LAB_6.ipynb 6.10
 recognise a known object LAB_5_6_LAB_TASKS.ipynb 6.3
 recognise in video LAB_5_6_LAB_TASKS.ipynb 6.4
 panorama / stitch LAB_5_6_LAB_TASKS.ipynb 6.5 (6-line
 version: 6.5b)
 intruder / zone / alarm LAB_5_6_LAB_TASKS.ipynb 6.8 (two stills:
 6.9)

WAVELETS (LAB 6)

DWT LAB_6.ipynb 3.1 / 3.2
 wavelet denoise image LAB_6.ipynb 3.4
 wavelet compression LAB_6.ipynb 3.5
 sensor / 1-D signal LAB_5_6_LAB_TASKS.ipynb 6.10
 2-D wavelet story LAB_5_6_LAB_TASKS.ipynb 6.11
 wavelet denoise + Canny + Hough .. LAB_6.ipynb 7.4

THEORY

Lab 1 MCQ LAB_1_3.ipynb 1.16
 Lab 2 and 3 MCQ LAB_1_3.ipynb 5.3 and
 LAB_1_3_LAB_TASKS.ipynb 5.4
 Lab 4 and 5 MCQ LAB_4_5.ipynb 5.3
 Lab 6 MCQ LAB_6.ipynb 8.1 to 8.4
 Lab 5 and 6 explain / why LAB_5_6_LAB_TASKS.ipynb 4.4